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Vortex-Induced Vibration Characteristics and Equivalent Static Force Calculation Method of Circular Steel Hangers on Arch Bridge

Xinyu Xu[a], Yingliang Wang[a], Xingyu Chen[a,b], Xiaolong Zheng[a], and Yongping Zeng[a,b]

[a]China Railway Eryuan Engineering Group Co., Ltd., Chengdu 610031, China
[b]Dept. of Bridge Engineering, Southwest Jiaotong University, Chengdu 610031, China

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ABSTRACT

In order to study the vortex-induced vibration (VIV) characteristics and equivalent static force calculation method of circular steel hangers on the arch bridge, a numerical simulation model of VIV was established using FLUENT software, and the reliability of the model was verified by wind tunnel tests. Effects of wind speed, damping ratio, initial tension and slenderness ratio on the VIV characteristics of circular steel hanger were investigated. Furthermore, the equivalent static force calculation formula and loading range of hanger considering VIV were presented. The results show that with the hanger's damping ratio decreasing, the maximum amplitude and the lock-in wind speed range increases gradually, while the wind speed and frequency of VIV at the maximum amplitude were not significantly affected. As the initial tension increases or the slenderness ratio decreases, the frequency of VIV increases, and the wind speed corresponding to the maximum amplitude increases, however the maximum vibration amplitude has few changes. Based on a number of numerical analyses on VIV characteristics of hangers, the equivalent static force calculation method is put forward. The equivalent static force calculation method could provide a reference for the design and static analysis of the circular steel hangers.

1. Introduction

In the modern bridge engineering, the hangers of arch bridges are mostly made of steel, which have the characteristics of light weight, large slenderness ratio and small damping ratio. With the increasing span of arch bridges and the increasing length of hangers of arch bridges, the vortex-induced resonance of the hanger is easy to occur at low wind speed, which may affect the driving comfort of vehicles, cause fatigue damage of hangers and even endanger the safety of bridges (Chen et al., 2013; An et al., 2017; Ma et al., 2018).

Railway bridges bear large loads and require high durability and large stiffness, therefore H-shaped or rectangular rigid hangers are usually adopted in railway arch bridges. However, large wind-induced vibration of these two types of hangers has been found in several projects, and vibration suppression measurements should be utilized to decrease the vibration of the hanger and to enhance the stability of the bridge even at a higher cost (Ulstrup, 1980; Uzgider et al., 2009).

Circular steel rigid hanger is a relatively new type of hanger in the railway arch bridge (Yoshimura et al., 2006; Ribeiro et al., 2012). However, with the length of hanger increasing, the wind-induced problems of the hangers are much more obvious (Vrouwenvelder and Hoeckman, 2004), while there is limited research on vortex-induced vibration (VIV) of circular steel rigid hangers. As for the circular or rectangular cross-section, there were several investigations on the flow and VIV characteristics by the numerical simulations and wind tunnel tests. The vortex dynamics of circular section were reviewed and some focuses were bought out (Williamson, 1996; Gabbai and Benaroya, 2005; Williamson and Govardhan, 2008). Zhao and Cheng (2014) studied the VIV of a rigid circular cylinder of finite length and found vortex shedding in the wake of a fixed cylinder is suppressed if the cylinder length is less than 2 cylinder diameters. Li et al. (2017) characterize the wind flow of the circular suspenders of suspension bridge in the wake of bridge towers at different wind directions by both numerical simulation and wind tunnel test. The VIV characteristics has been studied for the rectangles ($B/D \geq 4$) by

CORRESPONDENCE Xinyu Xu lsxxy90@126.com China Railway Eryuan Engineering Group Co., Ltd., Chengdu 610031, China

many researchers (Komatsu and Kobayashi, 1980; Matsumoto et al., 2008; Ricciardelli, 2010; Marra et al., 2015). A new generalized semi-empirical model for the rectangular section was proposed by Liu et al. (2018), which can simulate not only the interaction effects between VIV and galloping, but also the pure vortex resonance effects and pure galloping vibrations separately.

In the stage of bridge design, only the static wind load acting on the hanger is considered in the design of the hangers, but the load considering dynamic effects, like caused by VIV, is rarely considered. As for the hanger, the vortex-induced vibration is easy to occur at the frequent wind speed. At present, the additional force considering the VIV is not considered in the design code, and the equivalent static load calculation method of the hanger considering the VIV is not established as well. With the operating speed of train increasing, the stability of the bridge has been pain more attention. Hence, the study of the vortex-induced vibration characteristics of the circular rigid hanger and its equivalent static force calculation method is urgent to be conducted, to conveniently consider the VIV of the hanger due to the frequent wind in the design phase of the arch bridge.

In this study, the circular steel hanger of arch bridge is taken as the research object. Firstly, the wind-induced vibration model was established based on numerical simulation method and dynamic grid technology, and verified by comparing with the references. Then, the effects on the VIV characteristics of hanger were studied, such as wind speed, the damping ratio, the initial tension and the slenderness ratio. Finally, by formula deduction and data curve fitting, the equivalent static force calculation formula and the corresponding loading range were obtained. The calculation method could provide a reference for the static load analysis of the steel hangers.

2. Numerical Simulation Method and Model Verification of Vortex-Induced Vibration

2.1 Numerical Simulation Method

In this paper, Gambit was utilized to mesh the computational area of the numerical model, FLUENT was used to establish the VIV model, and Newmark-β method was adopted to simulate the dynamic characteristics of circular steel hanger. It should be noted that only the vertical vibration of the hanger was discussed in the study. The iteration program of the vertical vibration of the model was programmed with parameters, including the mass, damping and stiffness, via the program user-defined function (UDF) in FLUENT solver.

Firstly, the hanger was fixed and the numerical simulation calculation was conducted in the static state. In the calculation process, the shear stress transport (SST) k-ω model developed by Menter (1994) was used for the turbulence model. The SST k-ω turbulence model integrates the superiorities of the original k-ω in the near wall layer and the free stream independence of the standard k-ε model in the outer region. The pressure based solver and the implicit formulation were used, which were suitable for incompressible flow. The Semi-Implicit Method for Pressure-Linked Equations - Consistent (SIMPLEC) algorithm was applied to the pressure-velocity coupling, the second-order interpolation scheme was adopted for pressure, and the second-order upwind scheme was used for the moment and turbulence properties. The hanger would be released until the flow field was stable, and the transient lift of the hanger can be extracted, and the motion response of the hanger can be obtained (Chen et al., 2018; Li et al., 2018).

Through the dynamic grid technology, the numerical simulation

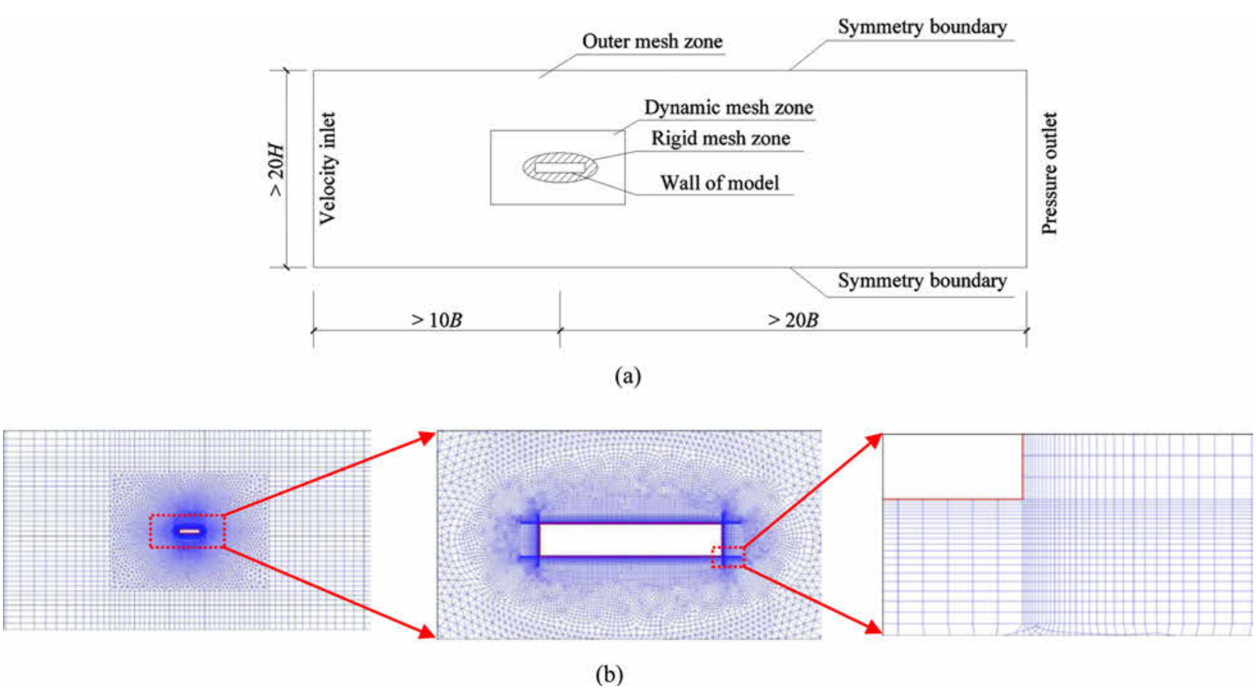

Fig. 1. Computational Area and Grid Mesh of Numerical Model: (a) Settings of Computational Area, (b) Diagram of Grid Mesh

of hanger motion can be realized, and the flow field is changed by the movement of the hanger. The change of flow field results in the change of lifting force acting on the hanger, which changes the motion response of the hanger in return. Thus, the vertical VIV simulation of the hanger can be realized through the cycle motion.

2.2 Model Verification

In order to verify the reliability of the numerical simulation method established in this paper, computational fluid dynamics (CFD) calculation of rectangular section was carried out, and the calculated results were compared with the experimental results. It should be emphasized that, compared with circular section, the rectangular section has distinct edges and corners, the phenomenon of fluid separation is more complete, so the requirements for grid quality will be more stringent. Thus, it shows that the grid generation method is feasible if the rectangle can be simulated accurately.

Firstly, to study the reliability of the numerical model in static state, the rectangular section was adopted in the calculation of the numerical model, with the size of 0.24 m × 0.04 m and the ratio of length to width of 1:6. The computational area and grid diagram of this section are shown in Fig. 1. There are 20 layers of the boundary layer, the height of the boundary layer grid was 0.5 mm, increasing by 1.1 times the height growth rate from inside to outside. In the rigid mesh zone, the grid outside the 20 layers of boundary layer still generates quadrilateral grid by 1.1 growth rate. In the dynamic mesh zone, triangle grid was generated by 1.1 growth rate. In the outer mesh zone, the height growth rate was set to 1.15. The mesh near the corner was locally encrypted, and the height width ratio of the mesh was controlled within 1:5. The wall y+ value of the whole rectangular shape was controlled within 2, and the y+ value of most positions was less than 1.

Among the model, the cross-section is loaded by drag force F_D, lift force F_L and moment F_M. In the wind axis coordinate system, the static three-component force coefficients of the section can be expressed as:

$$C_D = F_D / 0.5 \rho U^2 H \tag{1}$$

$$C_L = F_L / 0.5 \rho U^2 B \tag{2}$$

$$C_M = F_M / 0.5 \rho U^2 B^2 \tag{3}$$

Where,

B = length of the hanger parallel to the air flow
C_D = drag coefficient in the wind axis
C_L = lift coefficient in the wind axis
C_M = moment coefficient in the wind axis
H = length of the hanger perpendicular to the air flow
ρ = air density

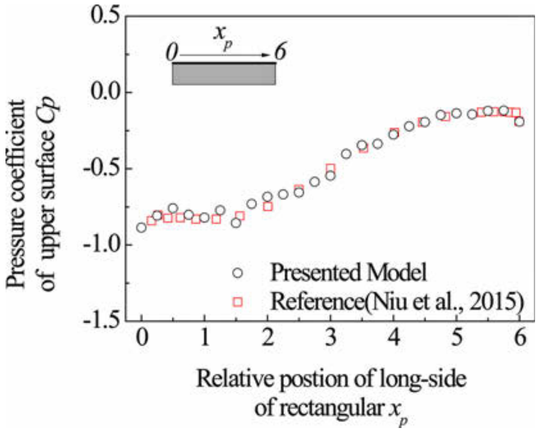

Fig. 2. Distribution of Surface Average Pressure Coefficient

Table 2. Parameters of Rectangular Section

Parameter	H/m	B/m	Mass/ kg	Frequency / Hz	Damping ratio ξ / %
Value	0.24	0.06	3.25	5.9	0.58

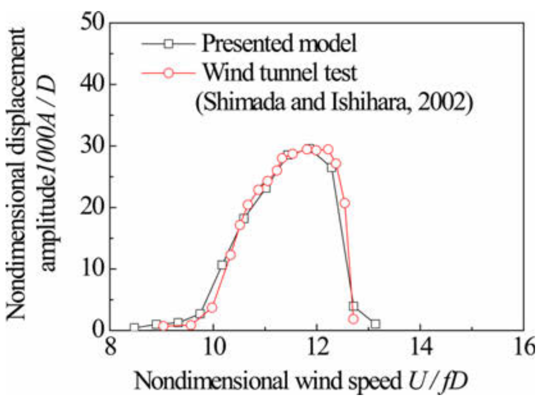

Fig. 3. Comparisons on the Displacement Amplitude Curves

U = the average wind speed of the incoming flow

Following the calculation procedure above, the aerodynamic coefficients were compared in Table 1, and the surface mean pressure coefficients ($C_p = p / 0.5 \rho U^2$) of the upper surface were compared in Fig. 2. In Table 1, the results of the presented model were close to those of CFD and wind tunnel tests in previous studies. And from Fig. 2, it can be found that the results from the numerical simulation are in good agreement with the previous research data, which demonstrating that the simulation accuracy meets the research requirements in the flow separation

Table 1. Aerodynamic Coefficients of Rectangular Sections

	Mean value of C_D	Amplitude of C_L	Strouhal number *St*	Reynolds number *Re*
Presented model	0.97	0.085	0.10	4,690
2D CFD (Niu et al., 2015)	0.99	0.088	0.10	4,690
Wind tunnel test (Niu et al., 2015)	—	—	0.11	4,690

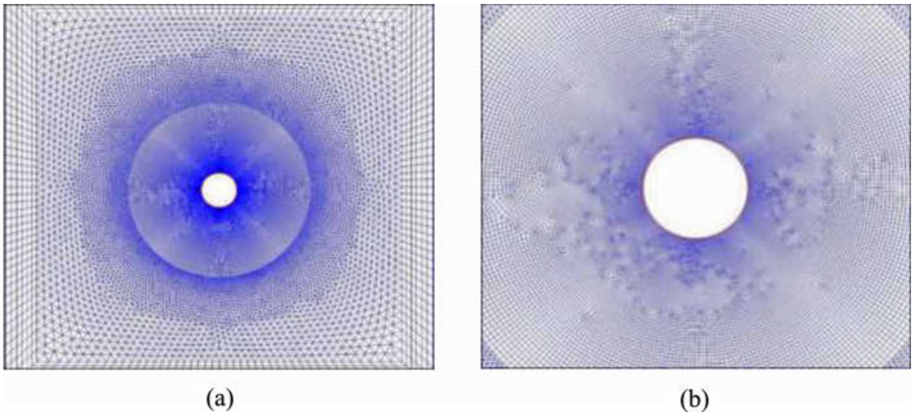

Fig. 4. Meshed Computatinal Zone: (a) General View, (b) Local View

area and reattachment area.

Furthermore, to verify the reliability of the VIV model, according to the parameters of the relevant research (Shimada and Ishihara, 2002), the parameters of the model were listed in Table 2. The results of numerical simulation were compared with those of wind tunnel test, which is shown in Fig. 3. The ranges of Reynolds number in Fig. 3 are both 12,000 – 18,600 for the presented model and wind tunnel test.

It can be seen that the numerical simulation method presented in the paper can simulate VIV. In the lock-in range, the relationship between the wind speed and the displacement were consistent with the wind tunnel test results.

The numerical simulation model established in the paper can reproduce the vortex-induced vibration well, and the simulation results are accurate and reliable.

3. Effects of Parameters of the Circular Steel Hanger on the Vortex-Induced Vibration

The whole computational domain was divided into several regions, including the rigid addition of refined rigid mesh area, dynamic mesh area and outer mesh area, as shown in Fig. 4. And the total mesh number is 42,270. The model has 20 layers of quadrilateral orthogonal body-fitted boundary layer grid. The y+

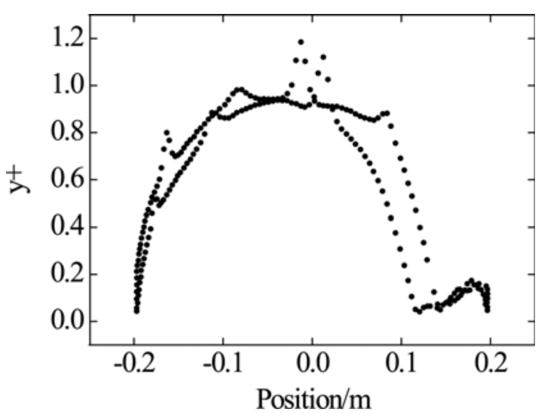

Fig. 5. y+ Value of Near-Wall Meshes of Circular Steel Hanger Model

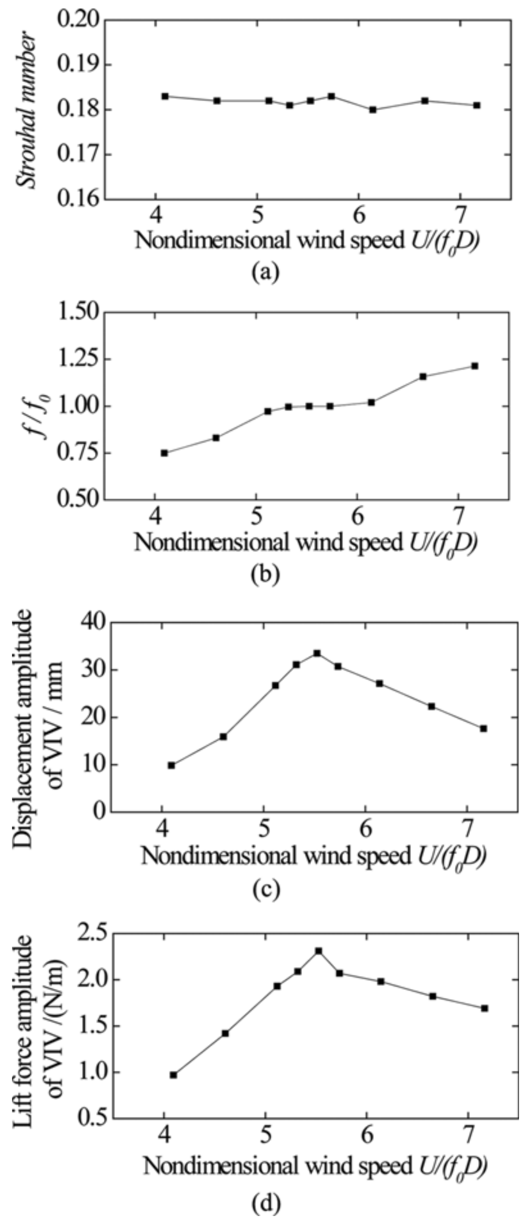

Fig. 6. VIV Responses of Circular Steel Hanger at Different Wind Speeds: (a) Strouhal Number, (b) Frequency Ratio of VIV, (c) Displacement Amplitude of VIV, (d) Lift Force Amplitude of VIV

value of the suspender near-wall mesh was shown in Fig. 5. It can be seen that the y+ value of the near-wall meshes was almost less than 1, which shows that the requirements of the mesh for SST k-ω turbulence model can be satisfied.

3.1 Effects of Wind Speed

In the study, the hanger in the practical arch bridge was taken as research project. The outer diameter of the hanger is 0.394 m, and mass per unit length of the hanger is 244.37 kg/m. According to the measurement, the structural damping ratio is chosen as 0.1%. The length of hanger is 35 m and the natural frequency f_0 is 2.48 Hz.

Obtained from the numerical simulation, the aerodynamic characteristics of circular steel hanger at various wind speeds was listed in Fig. 6, where the wind speed cases at the maximum amplitude are refined. It can be found that the Strouhal number of the circular steel hanger section does not change with the wind speed, and it is constant at about 0.18. The maximum displacement amplitude of VIV is 33.5 mm, occurring at nondimensional wind speed of 5.5. In the range of nondimensional wind speed of 5.3 – 5.7, the frequency of VIV is close to the natural frequency of hanger. Therefore, the VIV lock-in zone can be determined to be nondimensional wind speed range of 5.3 – 5.7.

The time histories of displacement and corresponding power spectral density (PSD) of the circular steel hanger at wind speeds of 4.0 m/s, 5.4 m/s and 7.0 m/s were displayed in Fig. 7, which corresponds to nondimensional wind speeds of 4.1, 5.5 and 7.1. It can be seen that the vibration of VIV shows the single frequency characteristics. When the maximum vortex-induced amplitude occurs, the frequency of VIV is equal to the natural frequency of the hanger.

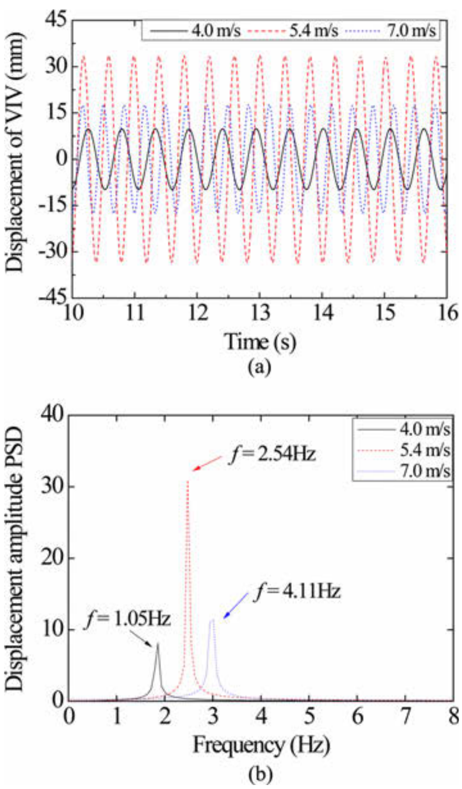

Fig. 7. Time Histories of Displacements of VIV and the Corresponding Power Spectral Density at Different Wind Speeds: (a) Time Histories, (b) Power Spectral Density

3.2 Effects of Damping Ratio

There are different regulations or suggestions on the damping ratio of steel structure in wind resistance codes of different countries or unions, the recommended values of damping ratio in different codes and some field tests are shown in Table 3. Previous studies have shown that the damping ratio of the structure has a significant impact on the wind-induced vibration of the tall towers or bridges (Li et al., 2003; Gu and Du, 2005).

In the CFD calculation, it is convenient to change the damping ratio of the structure. In this paper, the effects of damping ratio of 0.1%, 0.3% and 0.5% on the vortex-induced vibration were discussed. According to the Section 3.1, five representative wind speeds were utilized, including 4.0 m/s, 5.0 m/s, 5.4 m/s, 6.0 m/s and 7.0 m/s, which is in accord with the nondimensional wind speeds of 4.1, 5.1, 5.5, 6.1 and 7.1. The responses of VIV were shown in Fig. 8.

The results show that with the increase of damping ratio, the maximum amplitude decreases largely, and the lock-in range of VIV decreases as well. However, the change of damping ratio does not affect the corresponding wind speed of the maximum displacement amplitude. When the damping ratio increases from

Table 3. Damping Ratios for Steel Components in Different Codes and Some Field Tests

Source	Cable or hanger	Steel box girder	Steel truss girder	Steel tower
Wind-resistant design specification for highway bridges, China (JTG/T 3360-01-2018, 2018)	0.1%	0.3%	0.5%	0.5%
EUROCODE (EN 1991-1-4, 2005)	0.045%	0.2%	0.3% – 0.5%	0.5%
Design manual for roads and bridges - Loads for highway bridges (BD37/01, 2001)	—	0.2%		—
Wind resistant design of bridges in Japan — Developments and practices (Fujino et al., 2012)	0.03% – 0.2%	0.13% (Suspension bridge, average) 0.11% – 0.25% (Cable-stayed bridge)	0.22% (Suspension bridge, average)	0.13% – 0.3%
Long-term vibration tests of four steel hangers in Germany (Ruscheweyh et al., 1998)	0.2 – 0.5%	—	—	—

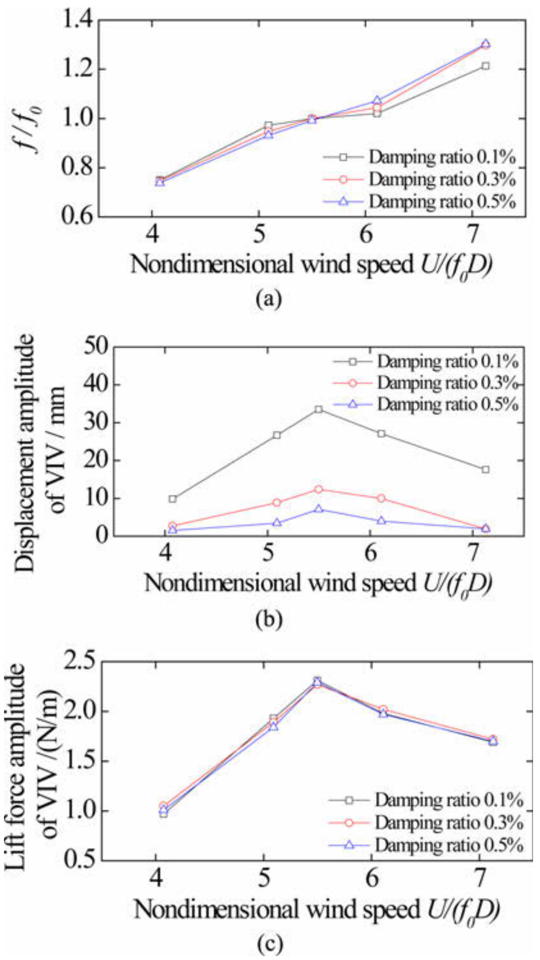

Fig. 8. VIV Responses of Circular Steel Hanger for Different Initial Tensions: (a) Frequency Ratio of VIV, (b) Displacement Amplitude of VIV, (c) Lift force Amplitude of VIV

0.1% to 0.5%, the displacement amplitude decreases rapidly from 33.5 mm to 7.1 mm.

3.3 Effects of Initial Tension

In practical engineering structures, to control the bending moment of the beam and to improve the performance of the rigid hanger, the initial tensions are applied on the rigid hanger on arch bridges. The initial tension will significantly affect the natural vibration frequency of hangers, and the vortex-induced vibration performance must be influenced further.

The natural frequency of bending mode of the tension-applied hanger can be calculated as follows:

$$f = \frac{22.4}{2\pi}\sqrt{\frac{EI}{ml^4}}\sqrt{1+\frac{N}{F_{eul}}} \tag{4}$$

Where,

E = the elastic modulus of the steel
I = the inertia of the cross-section
m = the mass of unit length of the hanger
l = the hanger length
N = initial tension
F_{eul} = the Euler critical load of the hanger

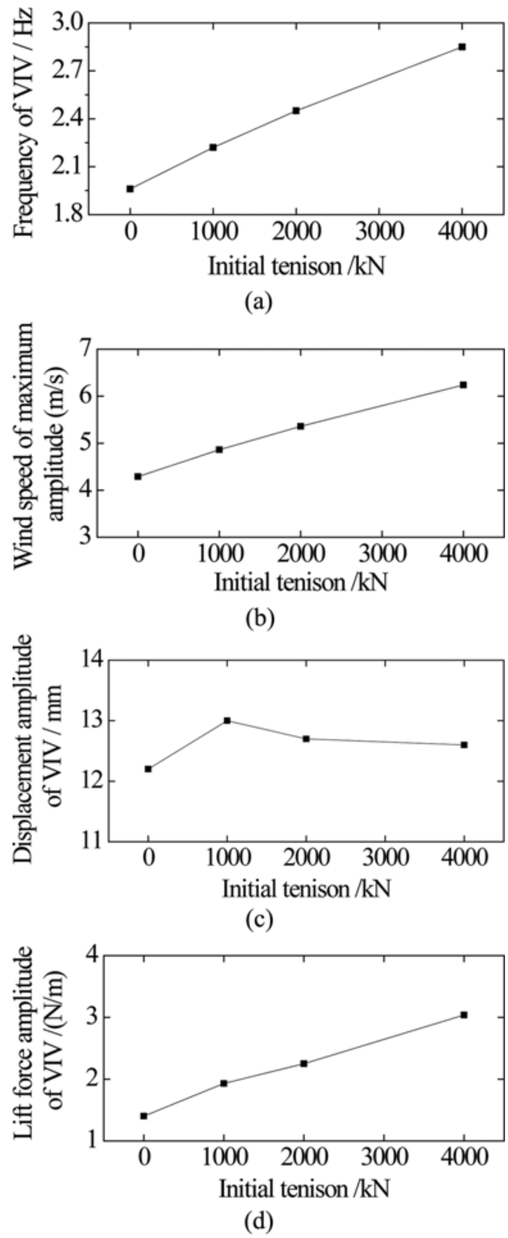

Fig. 9. VIV Responses of Circular Steel Hanger for Different Initial Tensions: (a) Frequency of VIV, (b) Wind Speed of Maximum Amplitude, (c) Displacement Amplitude of VIV, (d) Lift Force Amplitude of VIV

To study the effects of initial tension on the VIV of hanger, according to the research of the hanger tension (Kang et al., 2011; Xie et al., 2011) and relative experiences, four initial tension cases were calculated, including 0 kN, 1,000 kN, 2,000 kN and 4,000 kN. From the Eq. (4), the bending frequencies are 1.97 Hz, 2.22 Hz, 2.45 Hz and 2.86 Hz, respectively. According to the reference (Ruscheweyh et al., 1998), the damping ratio was chosen 0.3% in the following study.

The responses at the wind speed of maximum vortex-induced vibration amplitude were illustrated in Fig. 9. It can be seen that with the increase of initial tension, the frequency of VIV increase, which corresponds to the natural frequency, and the wind speed of maximum amplitude increases obviously and the lift amplitude increases as well. However, the initial tension has few effects on the displacement amplitude of VIV. Hence, at the certain wind speed, the hangers of different frequencies would probably vibrate in the same displacement amplitudes.

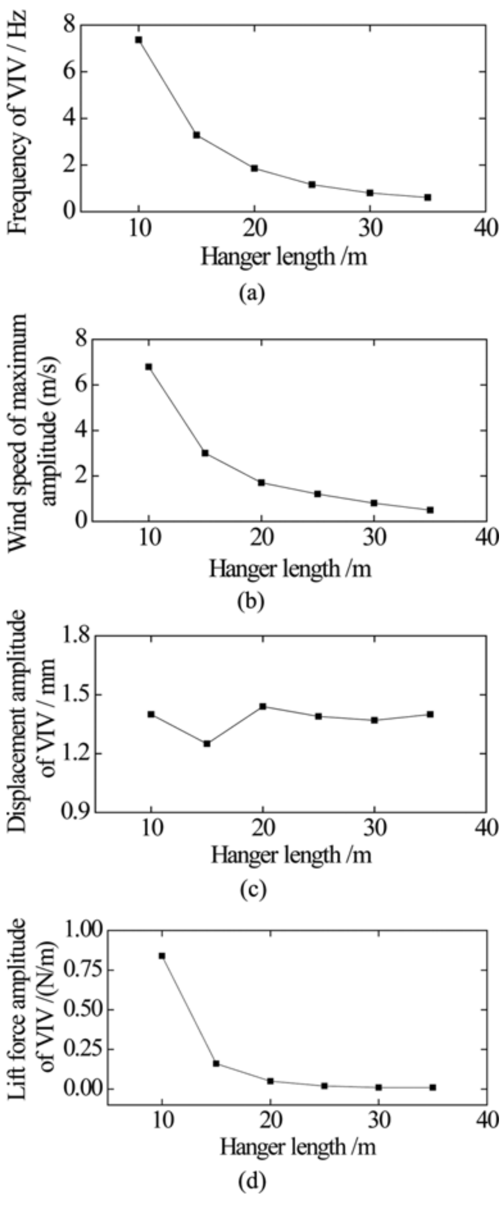

Fig. 10. VIV Responses of Circular Steel Hanger for Different Slenderness Ratios: (a) Frequency of VIV, (b) Wind Speed of maximum Amplitude, (c) Displacement Amplitude of VIV, (d) Lift Force Amplitude of VIV

Table 4. Frequency of the Hanger for Different Slenderness Ratios

	Hanger length / m					
	10	15	20	25	30	35
Slenderness ratio	62.5	93.75	125	156.25	187.5	218.75
Frequency / Hz	7.38	3.28	1.84	1.18	0.82	0.60

3.4 Effects of Slenderness Ratio

In the arch bridge, the dimensions of cross-section of hangers are usually the same, but the length varies greatly. To study the influence of slenderness ratio on the vortex-induced vibration, the length of the hanger adopted from 10 m to 30 m, and the diameter of the hanger was determined as 0.16 m. Thus, the slenderness ratio ranges from 62.5 to 218.75. The frequency for the hanger of different slenderness ratios were listed in Table 4.

Figure 10 shows the frequency, wind speed of maximum amplitude, displacement amplitude and lift force amplitude of VIV for different slenderness ratios. It is demonstrated that with the slenderness ratio of the hanger increasing, the wind speed of maximum amplitude and the lift force amplitude would decrease. However, there is no obvious changes for the displacement amplitude.

4. Equivalent Static Force Calculation Method

In the hanger design, only static wind loads are considered. Vortex-induced vibration belongs to the dynamic behaviors under wind action, but it is not taken into consideration in the hanger design. However, the studies above have already shown that the vortex-induced vibration of hanger is easy to happen at frequently occurring wind speed, and it would affect the performances of the hanger in operation.

To consider the vortex-induced vibration effects of the hanger in the static analysis, an equivalent static force calculation method was proposed. Firstly, through the theoretical deduction, the calculation formula of hanger considering structural dynamics was obtained. Then, the loading range of equivalent static force was obtained by comparing the static loading trial calculation with

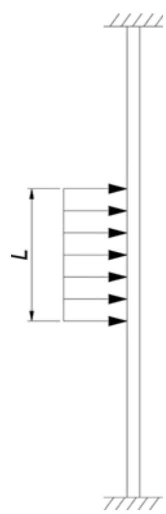

Fig. 11. Equivalent Static Force Loading Method

CFD simulation results. It should be noted that in the study the VIV of the hanger was analyzed considering the first-order symmetrical bending, for the hanger is more likely to vibrate for the lower frequency. The loading method is set as uniform distribution loading on a certain range around the hanger center, as shown in Fig. 11.

4.1 Equivalent Static Force Formula considering Vortex-Induced Vibration

Since the VIV of the hanger is basically the single-frequency and equal-amplitude motion, the lift force of the hanger in the lock-in region of the VIV can be described as a simple harmonic force. When the VIV occurs, the frequencies of lift force and structural motion are equal to the natural frequency of the structure. The displacement amplitude response of the hanger in the steady-state resonance can be expressed as:

$$y = F_L / 2k\xi \tag{5}$$

Where,

k = structural stiffness
y = displacement amplitude of the VIV
ξ = damping ratio

Considering the simplification of static loading, multiplying Eq. (5) by the stiffness k, substituting lift force of Eq. (2) and the Strouhal number $St = fD/U$, the equivalent static force can be formulated as:

$$\hat{F} = ky = \rho C_L \frac{1}{St^2}\frac{1}{4\xi} f^2 D^3 \tag{6}$$

Where,

D = the diameter of the circular hanger
f = structural frequency
$\hat{F}$ = equivalent static force considering VIV

According to Günther et al. (2000), when the natural frequency of the hanger is 7 Hz < f < 10 Hz, the vortex-induced vibration is not easy to occur. Hence, the frequency reduction factor k_f is considered. In addition, since the increase of wind speed would lead to the increase of turbulence in the natural circumstance, and turbulence around the structure would increase in the fully stable state, and the wind loads would decrease. Therefore, the wind speed reduction factor k_v is introduced (Ruscheweyh, 1982). The static force formula considering vortex-induced vibration can be formulated in the following equation:

$$\hat{F} = k_f k_v \rho C_L \frac{1}{St^2}\frac{1}{4\xi} f^2 D^3 \tag{7}$$

Where, if f < 7 Hz, k_f = 1; if 7 Hz ≤ f < 10 Hz, $k_f = (10 - f)/3$; and if the onset wind speed of VIV is less than 8 m/s, k_v = 1; if the onset wind speed of VIV is larger than 8 m/s, $k_v = (8/V_{cr})^3$.

4.2 Loading Range of Equivalent Static Force considering Vortex-Induced Vibration

Loading force can be obtained from Eq. (7). When the bending deformation at the center of the hanger is nearly equal to the displacement amplitude of the VIV, the loading length can be considered as the loading range of the equivalent static force.

Based on the results of CFD numerical simulation cases, the relationship between the loading range of equivalent static force was obtained through the trial calculation, and was fitted as follows:

$$L_{\hat{F}} / D = 0.4314 L / D - 2.36 \geq 5 \quad \text{solid circular steel hanger} \tag{8}$$

$$L_{\hat{F}} / D = 0.4314 L / D - 18.7 \geq 5 \quad \text{hollow circular steel hanger} \tag{9}$$

Where,

$L_{\hat{F}}$ = loading range of equivalent static force
L = the hanger length

5. Conclusions

The numerical simulation method of vortex-induced vibration was presented and verified in detail. Effects of wind speed, damping ratio, initial tension and slenderness ratio on the vortex-induced vibration of circular steel hangers were discussed. With the decrease of damping ratio, the maximum amplitude of VIV increases and the lock-in zone of VIV increases gradually. With the increase of initial tension or the decrease of slenderness ratio, the natural frequency of the hanger increases, and the wind speed corresponding to the maximum displacement increases. The change of initial tension and the slenderness ratio has no significant effects on the displacement amplitude. At the certain wind speed, the hangers of different frequencies are likely to vibrate in the same displacement amplitudes. Finally, the equivalent static force calculation formula and action range of hanger considering VIV were proposed. The proposed calculation method can be used in the design of circular rigid hanger, and also provide a reference for other types of hanger.

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ORCID

Xinyu Xu https://orcid.org/0000-0002-6850-5043

References

An YH, Guan DL, Ding YL, Ou JP (2017) Fast warning method for rigid hangers in a high-speed railway arch bridge using long-term monitoring data. *Journal of Performance of Constructed Facilities* 31(6):04017103, DOI: 10.1061/(ASCE)CF.1943-5509.0001097

BD37/01 (2001) Design manual for roads and bridges - Loads for highway bridges. BD37/01, British Standards Institution, Highways Agency, London, UK

Chen WL, Li H, Ou JP, Li FC (2013) Numerical simulation of vortex-induced vibrations of inclined cables under different wind profiles.

Journal of Bridge Engineering 18(1):42-53, DOI: 10.1061/(asce)be.1943-5592.0000323

Chen XY, Wang B, Zhu LD, Li YL (2018) Numerical study on surface distributed vortex-induced force on a flat-steel-box girder. *Engineering Applications of Computational Fluid Mechanics* 12(1):41-56, DOI: 10.1080/19942060.2017.1337593

EN 1991-1-4 (2005) Eurocode 1: Actions on structures — General actions — Part 1-4: Wind actions. EN 1991-1-4, European Committee for Standardisation

Fujino Y, Kimura K, Tanaka H (2012) Wind resistant design of bridges in Japan — Developments and practices. Springer, Tokyo, Japan

Gabbai RD, Benaroya H (2005) An overview of modeling and experiments of vortex-induced vibration of circular cylinders. *Journal of Sound and Vibration* 282(3-5):575-616, DOI: 10.1016/j.jsv.2004.04.017

Gu M, Du X (2005) Experimental investigation of rain — Wind-induced vibration of cables in cable-stayed bridges and its mitigation. *Journal of Wind Engineering and Industrial Aerodynamics* 93(1):79-95, DOI: 10.1016/j.jweia.2004.09.003

Günther GH, Hortmanns M, Schwarzkopf D, Bohmann D (2000) Dauerhafte ausführung von hängeranschlüssen an stählernen bogenbrücken. *Stahlbau* 69(11):894-908, DOI: 10.1002/stab.200003210

JTG/T 3360-01-2018 (2018) Wind-resistant design specification for highway bridges. JTG/T 3360-01-2018, Tongji University, Shanghai, China (in Chinese)

Kang HJ, Zhao YY, Zhu HP (2011) Analytical and experimental dynamic behavior of a new type of cable-arch bridge. *Journal of Constructional Steel Research* 101:385-394, DOI: 10.1016/j.jcsr.2014.06.005

Komatsu S, Kobayashi H (1980) Vortex-induced oscillation of bluff cylinders. *Journal of Wind Engineering and Industrial Aerodynamics* 6(3-4):335-362, DOI: 10.1016/0167-6105(80)90010-0

Li YL, Chen XY, Yu CJ, Togbenou K, Wang B, Zhu LD (2018) Effects of wind fairing angle on aerodynamic characteristics and dynamic responses of a streamlined trapezoidal box girder. *Journal of Wind Engineering and Industrial Aerodynamics* 177:69-78, DOI: 10.1016/j.jweia.2018.04.006

Li YL, Tang HJ, Lin QM, Chen XZ (2017) Vortex-induced vibration of suspenders in the wake of bridge tower by numerical simulation and wind tunnel test. *Journal of Wind Engineering and Industrial Aerodynamics* 164:164-173, DOI: 10.1016/j.jweia.2017.02.017

Li QS, Yang K, Wong CK, Jeary AP (2003) The effect of amplitude-dependent damping on wind-induced vibrations of a super tall building. *Journal of Wind Engineering and Industrial Aerodynamics* 91(9):1175-1198, DOI: 10.1016/S0167-6105(03)00080-1

Liu YZ, Ma CM, Li QS, Yan BW, Liao HL (2018) A new modeling approach for transversely oscillating square-section cylinders. *Journal of Fluids and Structures* 81:492-513, DOI: 10.1016/j.jfluidstructs.2018.05.014

Ma CM, Duan QS, Liao HL (2018) Experimental investigation on aerodynamic behavior of a long span cable-stayed bridge under construction. *KSCE Journal of Civil Engineering* 22(7):2492-2501, DOI: 10.1007/s12205-017-0402-7

Marra AM, Mannini C, Bartoli G (2015) Measurements and improved model of vortex-induced vibration for an elongated rectangular cylinder. *Journal of Wind Engineering and Industrial Aerodynamics* 147:358-367, DOI: 10.1016/j.jweia.2015.08.007

Matsumoto M, Yagi T, Tamaki H, Tsubota T (2008) Vortex-induced vibration and its effect on torsional flutter instability in the case of B/D = 4 rectangular cylinder. *Journal of Wind Engineering and Industrial Aerodynamics* 96(6-7):971-983, DOI: 10.1016/j.jweia.2007.06.023

Menter FR (1994) Two-equation eddy-viscosity turbulence models for engineering applications. *AIAA Journal* 32(8):1598-1605, DOI: 10.2514/3.12149

Niu HW, Zhou S, Chen ZQ, Hua XG (2015) An empirical model for amplitude prediction on VIV-galloping instability of rectangular cylinders. *Wind and Structures* 21(1):85-103, DOI: 10.12989/was.2015.21.1.085

Ribeiro D, Calçada R, Delgado R, Brehm M, Zabel V (2012) Finite element model updating of a bowstring-arch railway bridge based on experimental modal parameters. *Engineering Structures* 40:413-435, DOI: 10.1016/j.engstruct.2012.03.013

Ricciardelli F (2010) Effects of the vibration regime on the spanwise correlation of the aerodynamic forces on a 5:1 rectangular cylinder. *Journal of Wind Engineering and Industrial Aerodynamics* 98(4-5):215-225, DOI: 10.1016/j.jweia.2009.10.017

Ruscheweyh H (1982) Dynamische Windwirkung an bauwerken. Bauverlag, Berlin, Germany

Ruscheweyh H, Langer W, Verwiebe C (1998) Long-term full-scale measurements of wind induced vibrations of steel stacks. *Journal of Wind Engineering and Industrial Aerodynamics* 74-76:777-783, DOI: 10.1016/S0167-6105(98)00070-1

Shimada K, Ishihara T (2002) Application of a modified k-ε model to the prediction of aerodynamic characteristics of rectangular cross-section cylinders. *Journal of Fluids and Structures* 16(4):465-485, DOI: 10.1006/jfls.2001.0433

Ulstrup CC (1980) Aerodynamic lessons learned from individual bridge members. *Annals of the New York Academy of Sciences* 352(1):265-281, DOI: 10.1111/j.1749-6632.1980.tb16377.x

Uzgider E, Aydogan M, Caglayan O, Ozakgul K, Tezer O (2009) Performance of a tied-arch reinforced concrete railway bridge: Rating, safety assessment, and bond length evaluation. *Journal of Performance of Constructed Facilities* 23(5):366-371, DOI: 10.1061/(ASCE)CF.1943-5509.0000016

Vrouwenvelder ACWM, Hoeckman W (2004) Wind-induced vibration of tubular diagonals of the Werkspoorbridge. *Structural Engineering International* 14(4):314-321, DOI: 10.2749/101686604777963577

Williamson CHK (1996) Vortex dynamics in the cylinder wake. *Annual Review of Fluid Mechanics* 28(1):477-539, DOI: 10.1146/annurev.fl.28.010196.002401

Williamson CHK, Govardhan R (2008) A brief review of recent results in vortex-induced vibrations. *Journal of Wind Engineering and Industrial Aerodynamics* 96(6-7):713-735, DOI: 10.1016/j.jweia.2007.06.019

Xie X, Yuan P, Song YG (2011) A method for identification of short hanger tension considering damper. 2011 fourth international conference on intelligent computation technology and automation, March 28-29, Shenyang, China

Yoshimura M, Wu Q, Takahashi K, Nakamura K, Furukawa K (2006) Vibration analysis of the Second Saikai Bridge – A concrete filled tubular (CFT) arch bridge. *Journal of Sound and Vibration* 290(1-2):388-409, DOI: 10.1016/j.jsv.2005.04.004

Zhao M, Cheng L (2014) Vortex-induced vibration of a circular cylinder of finite length. *Physics of Fluids* 26(1):015111, DOI: 10.1063/1.4862548